

Luck in a Puzzling World: An Essay

Honggao Cao
August 2026

Abstract

Luck is easy to invoke after an outcome and difficult to identify independently of it. This essay examines what realized success and failure can reasonably tell us about strategy, capability, and merit. We distinguish the choices people make, the possibilities available to them, the events that occur, and the outcomes that follow. Concrete examples show why rarity and unexplained variation alone do not identify luck, and why a successful history does not automatically supply reliable advice for another setting. We then follow success beyond the first outcome: it can change the winner's future opportunities, while interpretations of success can spread and influence subsequent behavior. The central concern is greater discipline in attributing outcomes to merit, strategy, or luck, particularly when those attributions guide rewards, resource allocation, and the imitation of successful practices.

Keywords: luck; uncertainty; realization; success; attribution; transportability; cumulative advantage

1. Why Luck Bothers Us

People choose actions, but they do not choose outcomes.

The distinction sounds obvious until we try to explain success. A founder builds a company and becomes enormously wealthy. An investor follows a disciplined strategy for decades and accumulates a fortune. A scientist makes a discovery that changes a field. A young employee joins an obscure company shortly before it becomes an industry leader. Once we know how these stories end, the path leading to the outcome tends to become cleaner. The successful strategy becomes visible. The failed alternatives fade. Decisions that were made under uncertainty begin to look like steps toward a destination.

That is one reason luck is such an intriguing subject. We use the word easily, yet it becomes difficult to say exactly what we mean by it.

Consider a lottery winner. Few people object to saying that someone who buys a Powerball ticket and wins the jackpot was lucky. The person's action was necessary—without the ticket there is no prize—but buying the ticket plainly does not explain why that particular ticket won. The distinction between action and realization seems easy.

Now replace the lottery ticket with an investment strategy. Warren Buffett's extraordinary record plainly cannot be understood as a lottery drawing. Skill, judgment, discipline, institutional arrangements, capital, learning, and repeated decisions matter. But another question remains: how much does the extraordinary outcome itself tell us about how extraordinary each of those inputs must have been?

That question is harder than it first appears. If we begin with the winner and work backward, almost every feature of the winner can acquire explanatory significance. Patience becomes the reason for success. Concentration becomes the reason. Independence of thought becomes the reason. The ability to wait

becomes the reason. Each may genuinely matter. Yet the fact that a characteristic can be found in a successful person's history does not establish how much of the success it caused.

The problem is symmetric. A failed entrepreneur may have made poor decisions. Or a sensible strategy may have encountered an unfavorable realization. A scientist whose promising project produced nothing may have lacked insight, or the relevant evidence may simply not have appeared. A good decision can end badly; a bad decision can end well.

Experimental research shows how difficult it is for people to preserve this distinction after learning the result. Baron and Hershey found that people evaluated decisions and decision-makers more favorably when outcomes were favorable, even when the information available to the decision-maker was held constant [1]. The outcome reaches backward and changes our interpretation of the choice.

Selection makes the problem worse. We study winners because they won. Denrell showed how learning from surviving and successful organizations can generate false beliefs about effective management practices when failures are undersampled [2]. The missing cases are not merely an inconvenience. They can change the apparent relationship between a practice and success.

These studies give us a reason to examine how the outcome was produced before treating it as evidence of exceptional capability. An extraordinary outcome records what happened; explaining it requires evidence about the choices, circumstances, and events that led there.

That puzzle reaches well beyond famous investors and entrepreneurs. It appears whenever outcomes become evidence about merit. We promote managers whose projects succeeded. We allocate capital to investment managers with strong records. We imitate successful organizations. We ask accomplished people for advice. We assign credit for success and blame for failure. In each case, the observed outcome becomes information about the person and the strategy.

Sometimes that inference is justified. Sometimes it is not. The difficult part is knowing which case we are in. This is where a study of luck becomes more than an exercise in defining a familiar word. The subject sits at the boundary between what people do and what happens to them, between agency and uncertainty, and between a realized history and the alternatives that disappeared when that history became real.

We examine that difficulty by first asking what the familiar word luck describes. We then follow the path from a choice to its outcome, and from that outcome to the judgments and subsequent opportunities it creates. This allows us to keep the actor's contribution in view while examining what the realized history can reasonably establish.

2. What Do We Usually Mean by Luck?

The lottery gives us a useful starting point because its uncertainty is explicit. A person buys a ticket and wins a jackpot. The drawing selects one result from many possibilities, and the buyer does not control which result appears. Here luck seems closely related to chance.

Now consider a researcher who notices an unexpected result, recognizes its importance, and turns it into a discovery. Both cases involve something the person did not choose, but the researcher's response adds a different kind of contribution.

In the scientific case, the word luck alone leaves that contribution unclear. An unexpected observation may occur without being sought, but recognizing its value can require preparation, expertise, curiosity, and action. Research on serendipity has therefore distinguished it from generic luck. Recent synthesis work emphasizes surprise, agency, and value as central elements of serendipity [3]. The unexpected event matters, but so does what the person does with it.

A commuter who takes a different route one morning and meets someone who later offers a career-changing opportunity introduces another distinction. The encounter sounds like "right place, right time." Career research has studied such chance events directly: participants report unplanned events and encounters influencing career decisions and transitions [4–7]. In this case, timing and position matter even though no lottery-like device selects the encounter.

An unusually successful fund manager raises a more explicitly statistical question. Could performance this strong occur without exceptional investment skill? Researchers address that question by comparing observed performance with results that could arise under a specified benchmark model. Bootstrap studies use repeated resampling to construct such comparison distributions [8]. Here luck refers to sampling variation under the model, and the inference depends on the benchmark chosen.

Philosophy adds further distinctions. Accounts of luck have appealed to low probability, lack of control, significance to the affected person, and what philosophers sometimes call modal fragility—the idea that a lucky event could easily have gone otherwise in nearby possible circumstances [9,10]. The literature on moral luck asks a related but different question: how should responsibility be understood when outcomes depend partly on circumstances beyond an agent's control? [10]

These interpretations overlap, but none simply contains all the others.

Low probability, for example, seems promising until we shuffle a deck of cards. Every exact ordering of a properly shuffled deck is extraordinarily improbable. Yet we do not normally exclaim that the resulting sequence was extraordinarily lucky. Something more is needed—perhaps significance, prior specification, control, or consequence.

Lack of control is also too broad. Tomorrow's sunrise is outside an individual's control, but observing it is not ordinarily considered lucky. Conversely, a person may deliberately increase the probability of encountering favorable opportunities without controlling which opportunity actually appears. Career theories of planned happenstance make exactly this distinction: exploratory behavior can increase exposure to unplanned opportunities without turning those opportunities into predetermined outcomes [11].

"Right place, right time" has intuitive appeal, but it raises another question. Is that simply a convenient retrospective description? Once a career succeeds, it is easy to identify the meeting, location, institution, or historical moment that now appears decisive. How would we have identified the same event before knowing where the trajectory ended?

Even randomness is not enough. A random event can be trivial. A deterministic event can be experienced as fortunate if the person could not reasonably know or control it. And an event that is fortunate for one person may be unfortunate for another.

These examples show why a single familiar word can conceal different questions. We may be asking how an event was selected, how surprising it was, what the person controlled, how the person responded, or why the event mattered. Keeping those questions distinct gives us a better basis for examining a particular outcome.

The interpretations may share a deeper structure, or luck may gather several kinds of experience under one name. We can investigate those possibilities without settling the definition in advance. First, however, we need to examine some shortcuts that appear to separate skill from luck while leaving the underlying events unexplained.

3. Some Easy Answers That Are Not So Easy

One tempting shortcut is to call whatever a model cannot explain luck. Suppose we try to explain an outcome Y using observed characteristics X , such as intelligence, effort, education, capital, strategy, and experience. Let f represent the relationship specified by the model and ε the portion left unexplained:

$$Y = f(X) + \varepsilon$$

Calling ε luck would give the unexplained portion a name, but it would not identify what produced it.

Residuals can contain measurement error, omitted variables, misspecification, unstable relationships, sampling variation, and unobserved interactions. If a model leaves Warren Buffett's wealth largely unexplained, we would need further evidence to determine which of those limitations matters and what produced the wealth. The residual itself cannot answer those questions.

Empirical studies of investment performance make this point concrete. Researchers can ask whether unusually high returns are consistent with chance under a specified asset-pricing model. But conclusions about skill and luck can depend on the benchmark, the bootstrap procedure, the treatment of dependence across funds, and whether skill is allowed to vary over time [8,12,13]. "Luck" in such a model is not a newly discovered force. It is an inference conditional on an analytical structure.

Another shortcut runs in the opposite direction: if an outcome is repeated, perhaps luck disappears.

Repeated success certainly provides more information than a single success. A person who performs extraordinarily once may have experienced a favorable realization. Someone who performs extraordinarily over many independent opportunities gives us stronger reason to infer persistent capability. But repeated outcomes are not always independent. Success can change the conditions of the next round.

A successful investor gains more capital and access. A successful scientist gains reputation, collaborators, and resources. A successful founder gains networks and the ability to recruit. A celebrated author gets a larger audience for the next book. The next opportunity set is no longer the one that existed before the first success.

The sociology of science has long described such cumulative advantage through the "Matthew effect" [14]. Randomized field experiments have gone further by bestowing arbitrary initial advantages in live social systems and observing whether those advantages generated additional success. Across several settings, they did [15].

Repeated success can therefore contain at least two kinds of information at once. It can reveal persistent capability, and it can reflect feedback from earlier success.

A final shortcut is to reverse the argument: if outcomes cannot be cleanly attributed to skill, they must be luck.

That inference has the same weakness as naming the residual. A poor explanation of merit supplies evidence about the limits of our explanation; it supplies no positive evidence for a particular alternative. Otherwise the study becomes self-sealing: unexplained success, rare success, and divergence between apparently similar people would all confirm luck merely because our model failed to explain them.

To move beyond that circularity, we need to look more closely at what happens between the choice and the outcome. This is where differences in strategy, available opportunities, and realized events can begin to be examined separately.

4. What Happened Between the Choice and the Outcome?

Consider a simple investment decision. At the end of today, an investor chooses a strategy. Tomorrow markets may rise or fall, a company may announce news, financing may remain available or disappear, or nothing important may happen. The investor chooses how to prepare and act today, while tomorrow still contains several possibilities.

We can slow down the story by distinguishing four objects. Let $S(t-1)$ denote the strategy selected before period t . Let $\mathcal{X}(t)$ denote the set of possible states relevant to the investment during that period, $X(t)$ the state that actually occurs, and $Y(t)$ the resulting outcome. The sequence is:

$$S(t-1) \rightarrow \mathcal{X}(t) \rightarrow X(t) \rightarrow Y(t)$$

The arrows organize the stages we need to examine; they are not an estimated causal model. Strategy can affect which states are feasible, their probabilities, and what the investor can do when a state occurs. The realized state then contributes to the outcome. Other relevant conditions are included in the state description rather than assumed away.

Buying a lottery ticket creates eligibility for a prize. Developing expertise can create opportunities unavailable to an unprepared person. Moving to a city can change whom someone might meet. Holding cash during a market crisis can make a purchase feasible when financing is unavailable. In each case, action changes the possibilities without selecting every event that follows.

A sensible strategy may therefore improve the probability of a favorable outcome while leaving that probability below certainty. Evidence that a strategy improves outcomes across appropriately comparable opportunities is evidence about strategy quality. One favorable outcome supplies less information because it is only one realization from the possibilities that remained open.

This distinction gives agency and uncertainty their respective places. A person can deserve credit for preparing well and responding capably while benefiting from circumstances that preparation did not produce. Conversely, a poor strategy can encounter a favorable realization. Evaluating the decision requires attention to the alternatives and information available when it was made.

Career research offers an intuitive parallel. Planned-happenstance theory emphasizes exploration that increases exposure to unexpected opportunities [11]. Studies of career chance events examine how unplanned events influence decisions and transitions [4–7]. Scientific serendipity similarly combines an unexpected observation with the capability to recognize its importance and develop it [3]. The event and the response both belong in the explanation.

Timing adds another distinction. Imagine that you somehow know tomorrow’s closing stock price today, while a suitable trade is still possible. Now imagine receiving the same price information one minute after tomorrow’s market closes. The statement about the price is identical, but its usefulness for acting before that close has disappeared. What the investor knows matters partly because of when the investor knows it.

When we observe the eventual outcome, these stages can easily collapse into a direct story about strategy. The outcome alone leaves us to reconstruct which possibilities action created, which state occurred, and how the actor responded. That reconstruction is the missing middle in many accounts of success and failure. It gives us a place to investigate luck by examining what happened there, rather than assuming that whatever we have yet to explain must be luck.

5. The World Does Not Hold Still

The stock-price example locates information relative to an opportunity to act. Over longer periods, the opportunities themselves can change. Imagine two investors who follow the same rule: keep cash available and buy strong businesses when fear has pushed prices below reasonable estimates of value.

Suppose the first investor encounters a broad selloff, identifies a business whose prospects remain sound, buys it, and benefits when confidence returns. The second investor follows the same rule during a period in which no comparably attractive purchase becomes available. The second investor keeps waiting. Even if both exercise sound judgment, their records can differ because only one encounters the opportunity that makes the rule productive.

This is a hypothetical comparison, and judging the businesses’ prospects would itself require evidence. Its purpose is to show why a strategy cannot be evaluated entirely apart from the circumstances in which it operates. The first investor’s success may reflect both preparation and an opportunity that the second investor never receives. Moving the same rule through time can change what following it makes possible.

Time is therefore awkward to treat as merely another item in a list of explanatory factors. Resources, knowledge, relationships, and strategy exist at particular moments, and their consequences depend partly on what else is possible then. Using the state $X(t)$ and outcome $Y(t)$ introduced above, we can express that dependence as:

$$Y(t) = f_t(X(t))$$

The subscript t on f indicates that the relationship between a state and its outcome can change across periods. This is a conceptual reminder, not a fitted equation. The same measured resources may lead to different outcomes when market conditions, institutions, or opportunities have changed.

Career research encounters this issue in ordinary settings. Careers can be influenced by unplanned meetings, project assignments, institutional developments, and timely opportunities [6,7,16]. Exploration can increase exposure to such events, as planned-happenstance theory emphasizes, while leaving the particular opportunity uncertain [11]. Preparation and historical circumstances work together rather than providing competing explanations.

People also occupy different positions within the same period. Two similarly trained professionals may work in different organizations, networks, or flows of information. One is assigned to a project led by someone who later becomes a sponsor; another joins a project that is cancelled. We can call this situated position: the person's location within the social, institutional, and informational environment. Historical time identifies when the person acts; situated position helps identify which opportunities the person can reach.

These differences become important when a successful executive says, "This is what worked for me." Before turning that account into advice, we need to ask what must remain true for the practice to work for someone else. Causal-inference research calls the transfer of causal conclusions across populations and environments a problem of transportability [17]. A successful history gives us a starting case, while evidence about the relevant similarities and differences determines how far its lessons can travel.

The familiar emphasis on favorable timing, advantageous circumstances, and harmony among people captures the same intuition. What someone can accomplish depends partly on when action occurs, where the person is situated, and how the person is related to others. The harder task is to establish which differences mattered in a particular case. Naming the favorable circumstances after seeing success is easier than showing their contribution—and the evidence available to us may leave much of that contribution unresolved.

6. The Things We Cannot See

Imagine two sibling dancers performing the same choreography on the same stage at the same time. One hand rises slightly higher. One turn begins a fraction earlier. Their performances are not identical, and a director may notice one rather than the other.

The example creates a problem for any attempt to study luck by finding two "identical" people. There are no identical people, and no empirical record contains every causally relevant feature of two lives.

Suppose two people look similar in the characteristics we have recorded, yet their later outcomes differ greatly. That comparison establishes divergence relative to our measurements. Explaining the divergence requires more: we may have measured a characteristic badly, omitted a relationship or experience, missed an interaction, or used the wrong model. The recorded similarity does not make the underlying lives identical.

Historical cases impose an additional ceiling. We cannot rerun a person's life while holding everything fixed and changing one realization. The counterfactual investor who made the same decisions but encountered different markets is unavailable.

Selecting extraordinary cases creates a further problem. We choose someone because the observed outcome is exceptional, then try to infer an underlying capability from that same outcome. Exceptional performance may partly reflect unusually favorable variation, so an estimate that ignores the selection can overstate capability. Research on inference after selecting apparent winners formalizes this difficulty [21].

The problem cuts both ways. A billionaire's wealth does not by itself establish extraordinary merit. Failure to explain that wealth with our variables does not establish extraordinary luck either.

The appropriate conclusion follows the evidence. An unexplained difference gives us a question to investigate, not a finding of luck.

Positive evidence would identify something that happened between the prior strategy and the later outcome: a documented encounter, allocation, draw, timing difference, or institutional change. The event must be identified independently of the outcome's being exceptional, and its contribution must be assessed in the relevant setting. Documentation can establish that an encounter occurred; stronger evidence may be needed to establish how much difference it made.

This gives research a useful task even when a complete decomposition is unavailable. We can establish parts of a realized path and identify where attribution remains uncertain. The same discipline becomes especially important when a winner's account begins supplying lessons for other people.

7. When Winners Explain Why They Won

Imagine two decision-makers facing an uncertain problem with the same information. One receives a favorable outcome and the other an unfavorable one. Which made the better decision? Experiments on outcome bias show how easily the result enters our assessment of the reasoning: decision-makers and their decisions were evaluated more favorably when outcomes were favorable [1].

Now consider an entrepreneur whose company succeeds. Journalists ask what made the difference. The founder explains persistence, hiring, focus, customer attention, speed, or independence of thought. Each practice may have contributed. But learning from the account requires asking how the practice operated, what circumstances supported it, and what happened to other founders who followed a similar approach.

Those other histories are less visible. Unsuccessful founders who were equally persistent do not receive the same audience. Denrell's analysis shows how observing surviving organizations disproportionately can make practices appear associated with success even when the relationship differs in the full population [2]. Selection affects the lesson before anyone deliberately exaggerates it.

The missing middle can then disappear from the public account. A strategy is followed by success, so the strategy becomes the explanation. Success also becomes evidence of competence, and perceived competence gives the winner authority to advise others. The uncertainty present during the original decision has been replaced by confidence in a retrospective lesson.

Successful people may possess deep practical knowledge. The question is which parts of that knowledge the outcome demonstrates and which require further evidence. Experience of one realized path can reveal how a person responded to particular events. Applying that experience elsewhere also requires understanding whether the opportunities and relationships that supported the result remain available.

This is why authority gained from an outcome does not automatically confer predictive authority. Other people may act on the advice, so the gap between explaining one history and guiding another has consequences beyond our assessment of the winner. Moreover, success changes the winner's own circumstances. The next success may occur under conditions that the first success helped create.

8. When Success Starts Producing More Success

Many social systems do not reset after each round. A young scientist who receives an early award can gain visibility and resources. A founder whose first product succeeds may find it easier to raise capital or recruit. A successful investor controls a larger capital base and may obtain access unavailable to smaller investors. The first outcome changes the opportunities available for the next undertaking.

That feedback separates two questions often combined in a success story: how an advantage began and why it lasted. Merton's Matthew effect describes disproportionate recognition accruing to already eminent scientists [14]. Randomized field experiments went further by assigning initial advantages in live social systems and examining subsequent success. Across several settings, recipients of those advantages experienced higher rates of later success [15]. These results show that earlier success can itself become one cause of later success.

Related processes can operate at the level of competing technologies. In settings with increasing returns, early adoption can strengthen the attraction of further adoption. Arthur's analysis shows how small historical events can influence which technology gains an advantage and how that advantage can become entrenched [22]. The relevance here is the change in subsequent conditions: an early event can help create the environment in which later choices are made.

Repeated extraordinary performance may therefore combine persistent capability, learning, accumulated resources, reputation, access, and feedback. A later success does not simply repeat the first test under unchanged conditions. Understanding the record requires examining how those conditions evolved.

It helps to separate the formation, amplification, persistence, and transmission of an advantage. What produced the initial difference? What enlarged it? What kept it from disappearing? What allowed it to pass across organizations, networks, or generations? These are distinct questions rather than obligatory stages in a single sequence, and their answers may differ. An advantage can originate in one event, grow through access to resources, persist through institutional arrangements, and later benefit someone who had no role in its origin.

This distinction also tells us why the debate cannot be settled by deciding whether the first event was luck or skill. Even if we could settle that attribution, we would still need to explain what followed. Some consequences remain close to the successful person; others spread when people interpret and imitate the success.

9. When Success Stories Change the World

The effects of success extend through its audience. An investment strategy performs exceptionally well, and capital flows toward similar strategies. A company succeeds with an unusual practice, and competitors imitate it. Consultants package the practice and business schools teach it. What began as one history becomes a prescription for future action.

The interpretation now joins the process being interpreted. People observe an outcome, form an explanation, and change their behavior on the strength of that explanation. Their choices contribute to new outcomes. The accuracy of the original lesson matters because it helps direct these subsequent choices.

Social-learning research explains why this process can occur among people who are trying to make sensible decisions. Other people's actions can contain information worth using. Under some conditions, however, observing earlier choices produces informational cascades: people follow those choices despite their own information, and collective behavior can become fragile or settle on an inferior alternative [18]. A visible history of adoption may then appear more informative than it is.

Other research examines how interpretations acquire a wider audience. Management-fashion research studies the spread of practices through consultants, management writers, media, and professional institutions [19]. Narrative economics examines how circulating stories can influence expectations and economic behavior [20]. These perspectives help explain how a lesson from success can move beyond its original setting without each new user independently testing the lesson.

Feedback can make the resulting evidence difficult to read. Suppose an investment story attracts new buyers, whose purchases temporarily raise prices. The rise may appear to confirm the story, even though the buyers' response contributed to the apparent confirmation. This hypothetical example adds a reason to follow the history forward: an interpretation can influence the events later offered as evidence for it.

Luck is therefore more than a word attached to a completed event. How we attribute the outcome can affect who receives resources, whose advice is followed, and which practices spread. With those consequences in view, we can return to the question of what the word should help us identify.

10. So, Does Luck Exist?

The examples give us reason to take luck seriously, while leaving open whether the word identifies one common mechanism. A lottery drawing, an unexpected opportunity, and an advantage amplified through reputation each involve different relationships between action and circumstance.

What, then, connects these cases?

One established possibility is modal: an event is lucky when it could easily have gone otherwise. That captures an important intuition but raises difficult questions about which alternatives count as nearby [9].

Perhaps luck is relational: an event is lucky for someone because it matters to that person and because the realization was not determined by what that person knew, chose, or controlled beforehand.

Or perhaps luck is not one thing. The word may gather together stochastic realization, favorable circumstances, timing, positional advantage, chance encounters, unexpected opportunities, model-relative sampling variation, and ex-post attribution.

That possibility does not make the concept useless. It changes the research question from "What is the one mechanism of luck?" to "Which relationships among agency, realization, control, significance, and consequence are being described when we use the word?"

A useful starting point is to examine a realized event from the actor's position before it occurred: what the actor knew, what the actor could influence, which alternatives remained possible, and why the event mattered. This preserves the connection between agency and realization without treating every unexplained outcome as luck.

It gives inquiry a direction while leaving the definition open. The consequences for judging people and institutions will depend on what inquiry actually establishes.

11. Why the Answer Might Matter

Consider first how we evaluate an accomplished person. If repeated extraordinary outcomes remain strongly associated with persistent capability and strategy after context, selection, and feedback are addressed, the record would give us a stronger basis for inferring capability. Judgments about merit would still involve values, but they would rest on a clearer account of the person's contribution. Learning from the winner would also have a firmer empirical foundation.

If comparable strategies instead frequently lead to materially different realizations, outcomes would be noisier indicators of decision quality. Credit for success and blame for failure would need to account for that variability. The relevant difference is not whether success matters, but how much information it provides about what the person chose and did.

The same reasoning applies to persistent advantage. If small initial contingencies are strongly amplified by capital, reputation, networks, or institutions, explaining long-lasting inequality would require examining those mechanisms alongside its origin. If institutions substantially reduce the effects of initial contingencies, the evidence would instead help us understand mobility and resilience. Neither conclusion follows merely from calling an initial event lucky.

Finally, learning from success depends on whether the relevant relationships continue to hold elsewhere. If a strategy works only under particular historical or institutional conditions, benchmarking and imitation need to preserve those conditions or establish why the lesson remains applicable without them. Otherwise a sound account of one success can become unreliable advice for another undertaking.

These possibilities matter because societies routinely convert outcomes into judgments about merit, competence, deservingness, and expertise. Evidence about how outcomes arise can inform those judgments, even though it cannot by itself decide what rewards or obligations are fair. The practical task is to make the connection between the evidence and the judgment explicit.

12. What Can Research on Luck Realistically Accomplish?

Research on luck can do useful work without predicting who will get lucky next. It can examine the opportunities a strategy creates, the events that actually occur, and the actor's response. It can distinguish an initial advantage from the processes that enlarge or sustain it, and test how far lessons from success apply across people and periods.

The lottery winner and the accomplished investor remain different cases. In the lottery, we can identify a drawing that selects among eligible tickets. In an investment history, capability, choices, circumstances, and subsequent feedback may be much harder to separate. Giving both outcomes the same label does not remove that difference; recognizing the difference tells us what evidence each explanation requires.

This brings us back to the missing middle. Before interpreting an outcome as evidence of merit or luck, we need to ask what people chose, what could have happened, what actually happened, and what followed. We also need to examine how the explanation of that history became advice or changed subsequent opportunities. Where evidence answers only part of those questions, the attribution should remain correspondingly limited.

A precise allocation of every success between skill and fortune may be beyond what the evidence supports. Greater discipline in attribution is still attainable. It allows us to recognize good judgment without crediting it with every favorable circumstance, and to recognize uncertainty without making it an explanation for everything we cannot see.

People choose actions while outcomes remain open. Remembering that uncertainty after the result is known helps us judge the choice, learn from the experience, and resist turning a single successful history into a prescription for everyone else.

AI-Use Disclosure

Generative AI tools assisted with literature discovery, drafting, editorial review, and document preparation. The author directed the research, reviewed and revised the resulting material, and takes full responsibility for the paper's content.

References

- [1] Baron, J., & Hershey, J. C. (1988). Outcome bias in decision evaluation. *Journal of Personality and Social Psychology*, 54(4), 569–579. <https://doi.org/10.1037/0022-3514.54.4.569>
- [2] Denrell, J. (2003). Vicarious learning, undersampling of failure, and the myths of management. *Organization Science*, 14(3), 227–243. <https://doi.org/10.1287/orsc.14.2.227.15164>
- [3] Busch, C. (2024). Towards a theory of serendipity: A systematic review and conceptualization. *Journal of Management Studies*, 61(3), 1110–1151. <https://doi.org/10.1111/joms.12890>
- [4] Bright, J. E. H., Pryor, R. G. L., & Harpham, L. (2005). The role of chance events in career decision making. *Journal of Vocational Behavior*, 66(3), 561–576. <https://doi.org/10.1016/j.jvb.2004.05.001>

- [5] Bright, J. E. H., Pryor, R. G. L., Chan, E. W. M., & Rijanto, J. (2009). Chance events in career development: Influence, control and multiplicity. *Journal of Vocational Behavior*, 75(1), 14–25. <https://doi.org/10.1016/j.jvb.2009.02.007>
- [6] Hirschi, A. (2010). The role of chance events in the school-to-work transition: The influence of demographic, personality and career development variables. *Journal of Vocational Behavior*, 77(1), 39–49. <https://doi.org/10.1016/j.jvb.2010.02.002>
- [7] Kindsiko, E., & Baruch, Y. (2019). Careers of PhD graduates: The role of chance events and how to manage them. *Journal of Vocational Behavior*, 112, 122–140. <https://doi.org/10.1016/j.jvb.2019.01.010>
- [8] Cuthbertson, K., Nitzsche, D., & O'Sullivan, N. (2008). UK mutual fund performance: Skill or luck? *Journal of Empirical Finance*, 15(4), 613–634. <https://doi.org/10.1016/j.jempfin.2007.09.005>
- [9] Broncano-Berrocal, F. Luck. *Internet Encyclopedia of Philosophy*. <https://iep.utm.edu/luck/>
- [10] Nelkin, D. K. (2025). Moral luck. *Stanford Encyclopedia of Philosophy*. Substantive revision January 20, 2025. <https://plato.stanford.edu/entries/moral-luck/>
- [11] Mitchell, K. E., Levin, A. S., & Krumboltz, J. D. (1999). Planned happenstance: Constructing unexpected career opportunities. *Journal of Counseling & Development*, 77(2), 115–124. <https://doi.org/10.1002/j.1556-6676.1999.tb02431.x>
- [12] Fama, E. F., & French, K. R. (2010). Luck versus skill in the cross-section of mutual fund returns. *The Journal of Finance*, 65(5), 1915–1947. <https://doi.org/10.1111/j.1540-6261.2010.01598.x>
- [13] Harvey, C. R., & Liu, Y. (2022). Luck versus skill in the cross section of mutual fund returns: Reexamining the evidence. *The Journal of Finance*, 77(3), 1921–1966. <https://doi.org/10.1111/jofi.13123>
- [14] Merton, R. K. (1968). The Matthew effect in science. *Science*, 159(3810), 56–63. <https://doi.org/10.1126/science.159.3810.56>
- [15] van de Rijt, A., Kang, S. M., Restivo, M., & Patil, A. (2014). Field experiments of success-breeds-success dynamics. *Proceedings of the National Academy of Sciences*, 111(19), 6934–6939. <https://doi.org/10.1073/pnas.1316836111>
- [16] Barbulescu, R., Jonczyk, C., Galunic, C., & Bensaou, B. (2022). Management of fortuity: Workplace chance events and the career projections of up-or-out professionals. *Journal of Vocational Behavior*, 139, 103791. <https://doi.org/10.1016/j.jvb.2022.103791>
- [17] Pearl, J., & Bareinboim, E. (2014). External validity: From do-calculus to transportability across populations. *Statistical Science*, 29(4), 579–595. <https://doi.org/10.1214/14-STS486>
- [18] Bikhchandani, S., Hirshleifer, D., Tamuz, O., & Welch, I. (2024). Information cascades and social learning. *Journal of Economic Literature*, 62(3), 1040–1093. <https://doi.org/10.1257/jel.20241472>
- [19] Abrahamson, E. (1996). Management fashion. *Academy of Management Review*, 21(1), 254–285. <https://doi.org/10.5465/amr.1996.9602161572>
- [20] Shiller, R. J. (2017). Narrative economics. *American Economic Review*, 107(4), 967–1004. <https://doi.org/10.1257/aer.107.4.967>
- [21] Andrews, I., Kitagawa, T., & McCloskey, A. (2024). Inference on winners. *The Quarterly Journal of Economics*, 139(1), 305–358. <https://doi.org/10.1093/qje/qjad043>

- [22] Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394), 116–131. <https://doi.org/10.2307/2234208>